

Pattern-Chain Online Learning Model

Memory-based learning, prediction, and generation

Overview

This document summarizes a memory-chain learning algorithm discussed in the conversation. The model incrementally learns patterns from un-uniform data, chains them temporally, supports anomaly detection, prediction, and data generation without flattening inputs.

Core Idea

Input data is matched against stored pattern memories. The best-matching pattern is selected, updated, and temporally linked to previous patterns, forming a chain or graph of experience.

Pattern Matching

For input x_t and stored patterns w_k , activation is computed as $a_k(t) = w_k^T x_t$. The pattern index is selected using $\arg \max_k a_k(t)$.

Learning

Pattern memory is updated using local Hebbian learning. Temporal transitions between patterns are stored in a transition matrix T .

Prediction

Prediction is performed in pattern space using a linear transition: $s(t+1) = T^T s(t)$.

Generation

Generated data is reconstructed linearly from predicted pattern activations: $\hat{x} = W^T s(t+1)$.

Anomaly Detection

Anomaly detection is based on confidence, not worst match. If $\max_k a_k(t) < \text{threshold}$, the input is considered anomalous.

Difference Between max and arg max

max returns the confidence value of the best match, while arg max returns the index of the best-matching pattern.

Comparison with LSTMs

Unlike LSTMs, this model does not rely on backpropagation through time, avoids vanishing gradients, and stores explicit interpretable memory.

Comparison with Transformers

Transformers rely on global self-attention and dense embeddings, while this model uses explicit pattern memory, linear transitions, and local learning.

Comparison with Other Online Learning Models

Most online learning models update parameters to minimize a loss. This model accumulates explicit memories and learns structure incrementally with high interpretability.

Applications

Applications include anomaly detection, lifelong learning systems, edge devices, memory-augmented AI, image generation, and predictive modeling.

Key Takeaway

This is a memory-first, online, interpretable learning system rather than a traditional neural network.